Lecture 2: Introduction to Limits

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09/10/2026

Day 1 Recap

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2. How did the homework go? (Note: Length will not be as long for future HWs, this one just had a lot of topics to review)
3. Any questions, comments, concerns?

Quiz Time!

At this point it's quiz time!

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1. Get into neatly organized rows
2. Do not flip over the page and start until directed to
3. Should go without saying but no cheating!
    - No looking at neighbors papers
    - No calculators
    - No notes
    - No phones
    - etc. etc.
4. I'll write on the board the minutes you have left, crossing each number out when you have *less than* that number of minutes left to complete
5. When time is up, stop writing and hold your quiz up so I can grab it. If you finish early, just chill until the 5 mins are up.

Time to do some math!

Now let’s start the lecture!

Limits

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- It gives you information about the value that a function tends towards, as it's input tends towards some fixed input.
- It tells you what value the function is "approximating" around some fixed input.
- It tells you local information about the function around some *small neighborhood* of some fixed input.

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   (2) Using typical measurements like pH strips etc. we also can't get a precise measurement of the output variable either

How do we proceed if, for instance, we want to know the *exact* pH of a solution whose concentration of acid is *exactly* 0.5 moles per liter?

Approximation II

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Graphically:

Approximation III

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The Limit!

Limits I

Limit of a Function

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Explicitly: for any error tolerance $\epsilon > 0$, if we want $f(x)$ to be equal to L within an error of ϵ (e.g. $|f(x) - L| < \epsilon$),

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If no such value L exists

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If no such value L exists (i.e. the function *cannot* be well approximated around x_0) then we say that $\lim_{x \to x_0} f(x)$ *does not exist*.

Limits II

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Option 1 (easiest): Graphically!
Given a graph of a function, we can easily tell what value a function approaches just by examining where the graph tends towards as we near a given point.

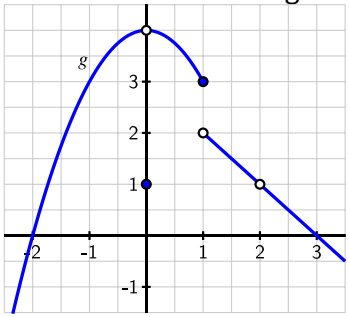


Figure: An example function

Limits III

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Option 2 (medium easy, *only an estimate!*): Evaluation!

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Given an expression of a function, e.g. $f(x) = x^2$ and a specific value around which to compute the limit, e.g. $x_0 = 2$,

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Given an expression of a function, e.g. $f(x) = x^2$ and a specific value around which to compute the limit, e.g. $x_0 = 2$, we can compute values that are closer and closer to x_0 *by hand/calculator*

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3. Group the h terms together to get $\lim_{h \to 0} 4 + h(2 + h)$
4. Notice that as $h \to 0$, $h(2 + h) \to 0$ as well so $\lim_{x \to 2} x^2 = 4$

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$$g(x) = \frac{x^2 - 1}{x + 1}. \tag{3}$$

Limits VII

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1. What is the domain of g?
2. For points x_0 in the domain of g, is it true that $\lim_{x \to x_0} g(x) = g(x_0)$?
3. What about the point that is *not* in the domain of g? What is $\lim_{x \to -1} g(x)$?

Limits VIII

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2. Does limit=value for points in the domain?
3. What about $x_0 = 0$?

Instantaneous Velocity I

What can we *do* with limits?

Average Velocity

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Suppose that an object's position as a function of time t is given by the function $s(t)$.

However, we rarely ever see *average* velocity in real life!

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$$AV_{[a,b]} := \frac{s(b) - s(a)}{b - a}. \tag{5}$$

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However, we rarely ever see *average* velocity in real life! E.g. if you drive a car then the dashboard reports your speed exactly in the instant you look at it, not over the last 1, 2, 5, etc. minutes!
(If it did, imagine someone driving slowly for a while, rapidly accelerating to a dangerous speed, getting pulled over and then trying to tell the cop "oh well my car said my average speed over the past 5 minutes was just 20mph" unlikely to avoid a ticket!)

Instantaneous Velocity II

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Suppose that an object's position as a function of time t is given by the function $s(t)$. The instantaneous velocity of the object at time $t = a$ is defined as

$$IV_{t=a} = \lim_{\Delta t \to 0} AV_{[a,a+\Delta t]} = \lim_{h \to 0} \frac{s(a+h) - s(a)}{h} \tag{6}$$

Instantaneous Velocity III

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1. What is the domain of $\dfrac{s(0+h) - s(0)}{h} = \dfrac{\sqrt{1+h} - 1}{h}$?

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1. What is the domain of $\dfrac{s(0+h) - s(0)}{h} = \dfrac{\sqrt{1+h} - 1}{h}$?
2. Can we determine the limit just by evaluating the above function at $h = 0$?

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1. What is the domain of $\dfrac{s(0+h) - s(0)}{h} = \dfrac{\sqrt{1+h} - 1}{h}$?
2. Can we determine the limit just by evaluating the above function at $h = 0$?
3. How could we get $\dfrac{\sqrt{1+h} - 1}{h}$ into a form where we *could* just evaluate?

Worksheet Time!

Now let's get into groups and work on the worksheet!

End of Class!

That’s all folks!